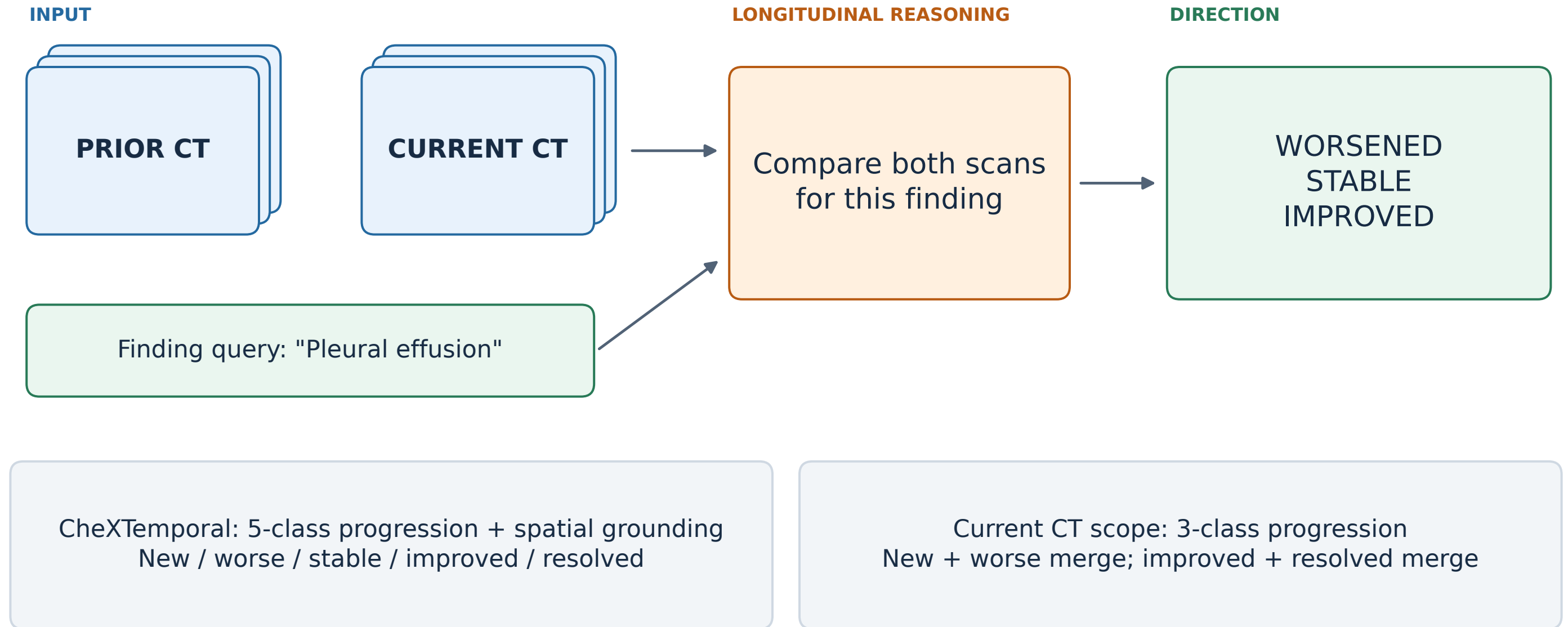


What changed between these CTs?

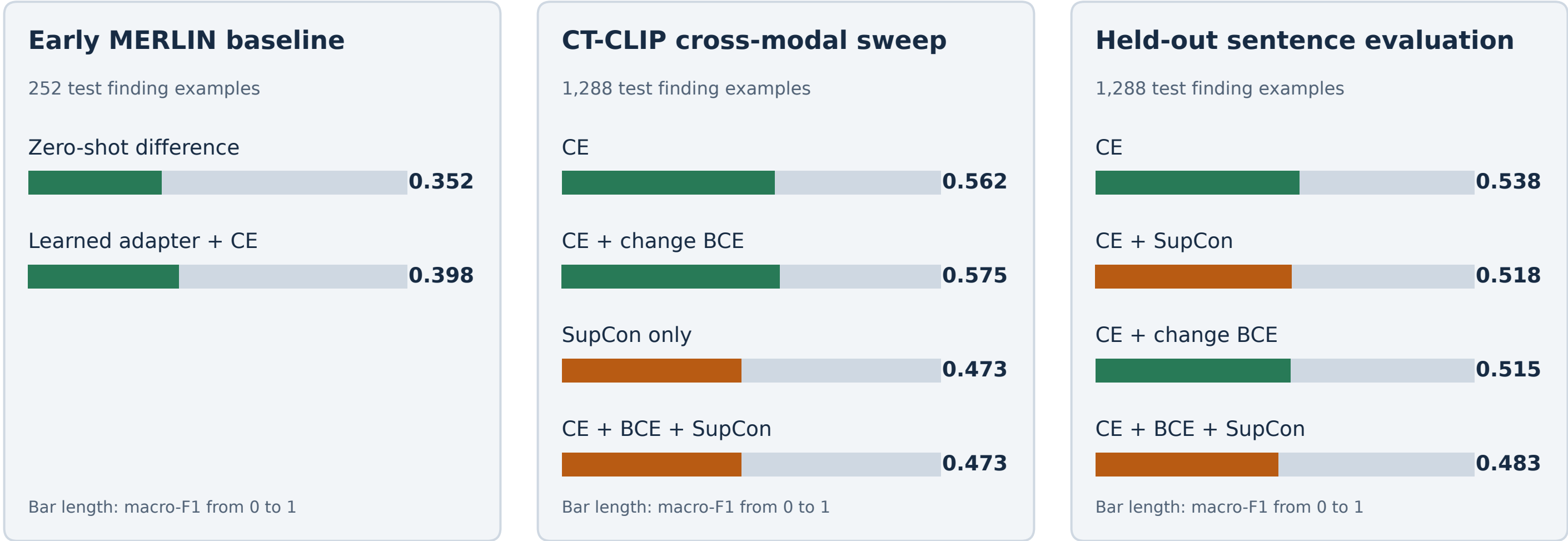
A CheXTemporal-inspired task: compare examinations, not just detect disease.



Spatial correspondence and localization remain future work. CT stacks shown schematically, not patient scans.

What the saved experiments show

Test macro-F1: descriptive results within each panel, not a cross-benchmark leaderboard.



Takeaway: CE-based models are competitive; adding SupCon did not consistently improve these runs.

Cohorts/labels differ across panels. SupCon sweeps also vary seeds and sampling; these do not isolate loss effects.

Middle: report-derived prototypes, 5,055 train examples. Right: held-out wording, 3 resamplings (not 3 training seeds).

Same task, different training signals

Example anchor: the learned CT-pair change embedding for "pleural effusion worsened".

Prototype cross-entropy

Choose the correct direction

CT-pair change embedding



Worsened prototype
Stable prototype
Improved prototype

Competes against 3 fixed text vectors for this finding.

Instance InfoNCE

Retrieve my paired sentence

CT-pair change embedding



+ Own report-text target
– Other batch text targets

Another correct same-class sentence can become a false negative.

Masked cross-modal SupCon

Match the correct language group

CT-pair change embedding



+ Same finding + same direction
– Same finding + other direction
Ignore other findings

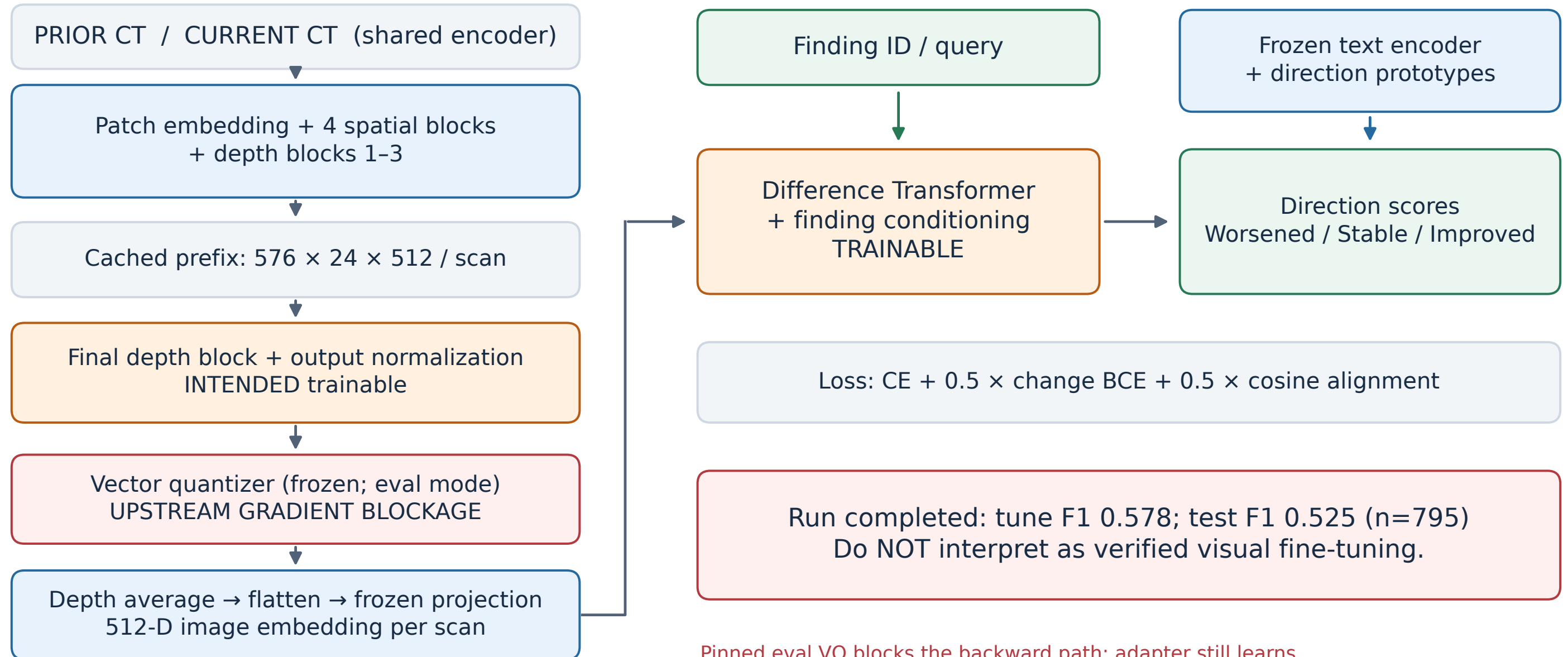
Multiple positives; compare only within this finding.

Last-block experiment: CE + change/stable BCE + paired cosine alignment (no SupCon).

Cosine alignment pulls toward one target without a contrastive denominator. BCE predicts changed vs stable, not physical size.

Last-block CT-CLIP: intended adaptation and actual status

Each scan is encoded separately; only the downstream Difference Transformer compares prior vs current.



Pinned eval VQ blocks the backward path; adapter still learns.

Fix gradient flow, check weight updates, then rerun. Reuse prefix cache.

Next: a controlled CT progression study

Change one factor at a time. Match examples, prototypes, seeds, sampling, and evaluation.

01

Validate the benchmark

Check labels + exact prior-study correspondence; review an image-based gold subset.

02

Prove both scans matter

Compare finding-only, current-only, prior-only, subtraction, and a paired adapter.

03

Isolate the training signal

Matched CE / change BCE / cosine / SupCon runs; choose settings on tune data.

04

Verify image fine-tuning

Repair VQ gradient path; confirm nonzero gradients and actual visual-weight updates.

Later: EXACT backbone comparison
Account for preprocessing and text-space differences.

Later: spatial grounding + acquisition robustness
Audit scanner/kernel effects before adding BiasCon.

Final evidence: patient-level uncertainty, per-finding results, and preferably an additional untouched evaluation set.

CT-RATE: from chest CT + reports to temporal inputs

Two preparation tracks: build valid longitudinal examples, then standardize each scan for CT-CLIP.

THE SOURCE DATA

Non-contrast 3D chest CTs Radiology reports + metadata

Local audit: 50,188 volume/report rows
21,304 patients
Multiple reconstructions per study

OUR TEMPORAL DATASET CONSTRUCTION

- 1 Choose one reconstruction per study; order by StudyDate.
- 2 Pair consecutive studies; drop same-day / missing-date pairs.
- 3 Use Findings + Impressions for report-derived direction labels.
- 4 Keep eligible findings; split by patient into train / tune / test.

CT-CLIP VOLUME PREPROCESSING • APPLY INDEPENDENTLY TO PRIOR AND CURRENT

LOAD NIFTI

Read intensities
+ voxel spacing
from the header

WINDOW + SCALE

Clip: -1000 to $+200$ HU
Linearly normalize
to $[-1, 1]$

RESAMPLE

$0.75 \times 0.75 \times 1.5$ mm
(x / y / z)
Linear interpolation

CENTER CROP / PAD

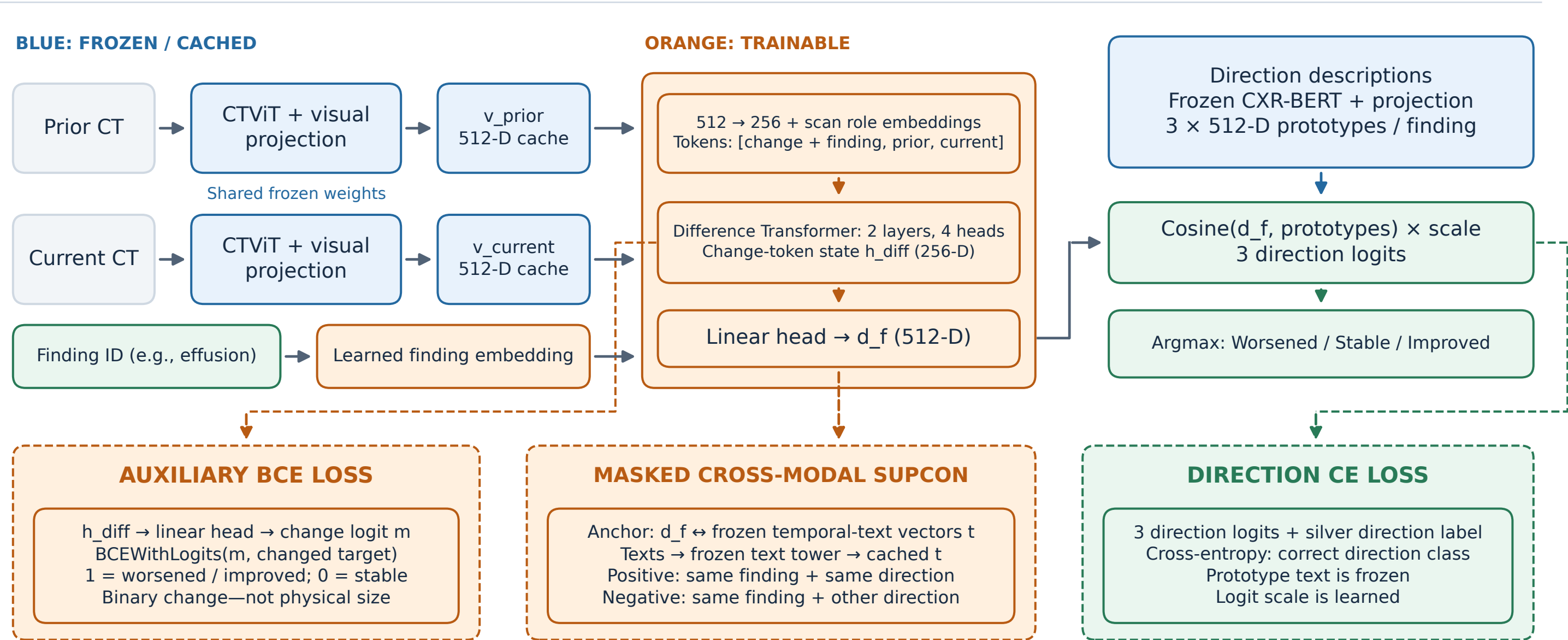
$480 \times 480 \times 240$
(H / W / D)
Pad value: -1

Input: (B, 1, 240, 480, 480) | Frozen adapter: cache 512-D vectors | Last-block: cache prefixes

Silver labels, not image-verified gold. No scan registration; exact report-to-prior correspondence still needs verification.

Frozen CT-CLIP + trainable temporal adapter

Frozen scan features → finding-conditioned comparison → direction scores. Dashed branches are training only.



Loss ablations: $\lambda_{CE} L_{CE} + \lambda_{BCE} L_{BCE} + \lambda_{SupCon} L_{SupCon}$. Terms switch on/off by run; SupCon ignores other findings.

Inference uses neither BCE nor SupCon, and needs no patient report. All CT-CLIP weights stay frozen; only the adapter/task parameters learn.